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Executive Summary

To the minister of State for Housing and planning:

As the world's socioeconomic disparities increase, more people face the ever looming threat of homelessness every year. In the last year, the British government has spent £1.7 billion on the fight against homelessness. Brighton and Manchester represent two fundamentally varying cities. Manchester is a larger city, with higher unemployment rates, more expensive land because of its population density and a more limited housing supply. Whereas, Brighton reflects a gentrified city which gatekeeps the poor from entering the housing market. ^[20] Therefore, its essential for you to understand how different strategies: namely controlling the population via immigration policy; increasing monetary aid/minimum wage; and increasing the number of unoccupied houses by construction, can combat homelessness.

We extrapolated the linear trend of the housing supply of the cities over time, allowing us to predict the housing supply of the cities in the future using a linear regression model. To model homelessness, we considered three dominant factors: money, housing and population. We first proposed and verified functions then investigated them individually, allowing us to accurately combine them into a single function. This let us not only predict values but investigate how the function behaves e.g. in its growth. This was pivotal in

We decided on determining the significance of three main factors: Population, Number of houses and pounds spend per homeless person using the Granger causality test. We compared each of these three factors with the amount of homeless people. Which then we use the Granger causality test to look at how each of these factors correlate to the amount of homeless people. We found that each factor could reasonably forecast the amount of homeless people, however we found that with creating a function incorporating all of those, we were able to create a much more accurate and sophisticated function, which would then also allow us to extrapolate the function for future values. This was considered a reasonable conclusion looking at the p-value and the F-statistic, which we used to test for significance with an alpha value of 0.10. In addition, we looked at if the correlation would change whether looking at long or short term. For this we used three values (10, 20 and 50 years) and performed a Granger causality test for all of them. We found that the pounds spend per homeless person had the biggest correlation.

Finally, we created an ideal model to address homelessness based on the data collected and our answer to question 2. We define the success in addressing homelessness as limiting the cost of the negative externality of homelessness on the community. Up until a certain threshold, homelessness costs more to the city than benefits. This model while explore the equilibrium between these two functions. Thus, the greatest amount of money saved will allow the community to invest in renewable energy or education. The strength of our model is its depoliticization. It effectively lets the *invisible hand* to set the equilibrium, thus making it

certain that utility will be maximized, especially in times of crises (migratory, financial, natural) where the pitfalls created by unbalanced policies – of both austerity and overspending – can be avoided. Throughout we used a mixture of percentage error, R squared values and RMSE to compare values, primarily RMSE when predicting and R squared if otherwise. [16].

We believe that these results will assist policymakers in ensuring that investment will see the greatest results in reducing homelessness in HIC cities of similar structure, not just those in the UK.

Global assumptions

A – 1. For our purposes, homeless people are considered as earning less money than non-homeless people.

- **Justification:** People's economic status is linked to their earnings and whether they have a stable home. Homelessness, indicating unstable housing, often means lower income due to factors like joblessness or low wages. Thus, homeless people are seen as earning less than those with homes, as their financial struggles affect their income [1].

A – 2. People can move freely within a city.

- **Justification:** Brighton and Hove, and Manchester offer extensive public transport, including buses, trains, and trams, enabling easy city-wide movement. Including free buses through Manchester, and Brighton and Hove buses make it easy for people to travel around the city and are cost effective [2].

A – 3. We consider a homeless person as someone that has no home in the UK or anywhere else in the world, that is available for them to live in.

- **Justification:** While the city council of Manchester does not provide a clear definition, it is the definition used by the city council of Brighton and Hove [3].

A – 4. We assume that aids are distributed evenly to all homeless people regardless of whether they have priority needs or not.

- **Justification:** It keeps the model relevant despite changes of policy regarding the definition of priority.

1. Q1: It Was the Best of Times

a. Defining the problem

The problem asks us to create a model that predicts changes in the housing supply in Manchester, Brighton and Hove in 2033, 2043, and 2073, while indicating our level of confidence in our predictions. Our model will take into consideration the previous building data from the two cities, as well as plans from the city councils.

b. Assumptions

A – 1. City councils will achieve their predicted number of houses built.

Justification: Brighton and Hove have backed their housing strategy with an independent feasibility review [4]. Manchester’s housing strategy is underscored by a range of projects that will support the delivery of it [5]. Both are bound by the accountability of the city councils to the local populations.

A – 2. The increase in number of housing units versus time in years is going to be approximately linear.

- **Justification:** For simplicity’s sake, we are assuming that the number of housing units over time shows a constant, linear increase. This simplifies the mathematical analysis, making it more applicable for our model.
- Although there might be unexpected changes in the number of housing units due to economic conditions, we are not considering these anomalies because the long-term trend caused the increasing demand for houses and the continued immigration which will cause the population of these cities to grow in the next 50 years.

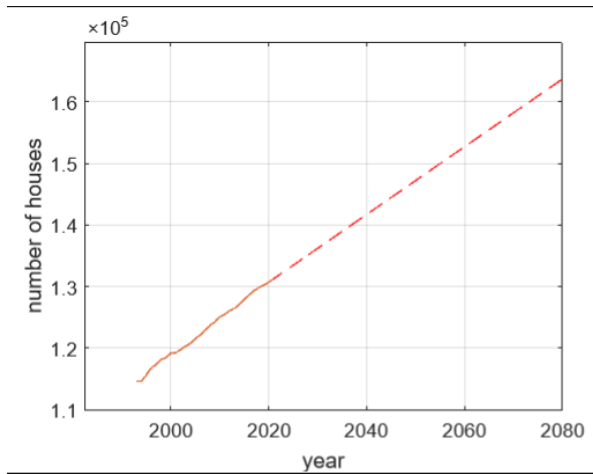
A – 3. Exogenous factors, such as the Covid-19 pandemic or the war in Ukraine will not affect the house-building process trend.

2. **Justification:** The effects of the Covid-19 pandemic on UK housebuilding industry have died down as lockdown restrictions are completely lifted [6]. The war in Ukraine has had no impact on the house-building industry so far [7].

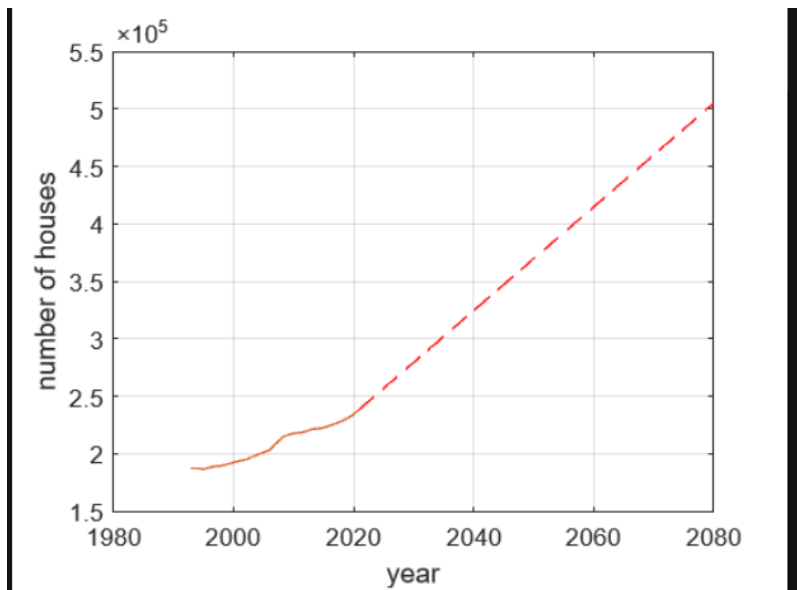
c. Variables

Symbol	Definition	Units
N	Number of houses available in a given city, per year	Units/year
T	Year	Time

d. The Model



Brighton & Hove



Manchester

e. Results

	2034	2044	2074
Number of houses in Brighton and Hove (millions)	1.375	1.425	1.59
Number of houses in Manchester (millions)	3.1	3.45	4.6

f. Discussion

In summary, our models predict that the number of dwellings will continue to increase linearly – in Brighton and Hove to 1.375 million in 2034, to 1.425 million in 2044, and to 1.59 million in 2074; in Manchester to 3.1 million in 2034, to 3.45 million in 2044, and to 4.6 in 2074. This is due to a fundamental demand for houses, epitomized by the 10-year pledge of both Labour and Conservative parties to deliver a growing number of dwellings [8]. We have considered that two of the underlying reasons for this linear curve is the increasing demand in the past years and the continued immigration which will cause the population of these cities to grow for the next 50 years, if not for crisis/disaster.

g. Sensitivity analysis

We performed a sensitivity analysis using percentage errors comparing the predicted population of Brighton and Hove's in 2025 of our models to the planned amount of housing in Brighton and Hove's and Manchester planned by the city council. In this Year, the predicted number from the city council was 134,520. In that year, our Model predicted 133,640 rounded to the nearest 10 houses.

$$P.E. = \frac{|predicted - actual|}{actual} \times 100\%$$

Using this formula, we get an absolute percentage error of 0.7%.

This means that our model is close to the actual amount of housing units in Brighton and Hove's, and Manchester and to some extent, it can accurately predict future housing supply.

By back testing the predictive linear models and comparing that to know-values, we obtained an RMSE value of 0.09 for Manchester, and 0.14 for Brighton and Hove. This reflects the nonlinear trend in much of the early years in Brighton which was caused by the 2008-09 global financial crisis. This event seems to have only delayed the growth in number of houses.

h. Strengths and Weaknesses

Using the model helps us determine the proximate long-term values of housing supply and dwellings, as extrapolation can be applied to our model because the data exhibits a clear increasing trend. The use of a simple linear model takes advantage of the simplicity of the expansion of housing planned by the city. It gives a simple overview of the number of housing units with reliable (see sensitivity analysis) data. Additionally, given that our data exhibits a clear linear increasing trend and that it does so for a reasonably long time of the coming 10+ years, we get a more precise overview of the problem at hand, without exceeding our own computational ability to access the complexity of the problem.

However, our model might not be completely accurate just because it assumes a linear relationship between number of house units and time. As can be seen from the graph from Manchester, our model has a limited ability to consider outliers caused by extraordinary economic events such as the financial crisis in 2008. Although we assume these events to not have a significant impact on the validity of our data, in real life, it will affect the number of housing units if not returning to the linear curve afterwards. Also, our model is only able to

determine values in the long term to a certain extent as we have based it on historical data whose accuracy is limited by human bias. These weaknesses reduce our confidence in the level of accuracy we have in our predictions of housing supply.

2. Q2: It Was the Worst of Times

For the regions you chose in Q1, predict changes in the homeless population in the next 10, 20, and 50 years.

a. Recomposing the problem

We are asked to model homelessness as time changes in Manchester and Brighton & Hove. We propose four dominant factors: annual salary, population number, housing supply, government spending.

We chose to focus mainly on these variables, as they represent comprehensively the problem, addressing it from the point of view of demographics (population number), economic strength (annual salary), governmental role (government spending), and housing market (housing supply). At first, we thought that we should consider economic strength via unemployment rates, especially as this would let us zoom into the affected group rather than having to use the properties of distributive techniques.

b. Assumptions

A – 1. We have considered the national poverty line as defined as the minimum income households require to reach a socially acceptable living standard. We have considered household which have income below 60% of the median household after housing cost as being below the national poverty line.

- **Justification:** This is the definition used by the United Kingdom government [8].

A – 2. Correlations between the number of homeless people and a potential underlying factor will be completely independent of other factors.

- **Justification:** When testing one of our factors against the number of homeless people, we will have to assume independence among other factors in order to run a Granger causality test. From there, we can determine the significance of any factors that may have a correlation.

A – 3. The GDP/capita of a certain area can be used as a representative measure of economic strength in a given city.

- **Justification:** Gross Domestic Product per capita is calculated by taking income earned from all sources and dividing it by the total population of a given area. It is often used as

a benchmark for improvements in the average real living standards of citizens from a certain area. Thus, we will use this value for our economic strength factor.

c. Variables

Symbol	Definition	Units
H	Number of homeless in a given city, per year	
N	Number of dwellings in a given city, per year.	
P	Number of inhabitants in a given city, per year	
I	Mean annual salary of homeless in a given city, per year	£/year
s_T	Total spending in nominal GBP on anti-homelessness programs, per year.	£/year
s_H	s_T/H GBP of aid per homeless person	£/year
H_p	Percentage of the population of a city that's homeless.	%

d. The Model

Model Development

We can reduce the number of inputs to 3:

$$s_H = \frac{s_T}{H} + I$$

Note that we can exclude the I term from our graphs as we are concerned only with the gradient and R^2 values from the monovariate inputs. This is because the I term operates as a translation which does not affect the gradient or the strength of correlation. In fact, by considering the I term, we risk increasing the uncertainty in our R^2 values as we would have to

$$\begin{aligned} \text{num houses available} &= N - \text{in use houses} \\ &= \text{houses} - \frac{P - H = \text{housed population}}{\text{mean household size} = 2.39} \quad [3] \end{aligned}$$

$$\text{num homeless} = \text{Function}(s_H, \text{Population}, \text{num houses available})$$

This lets us reduce the problem to three variables.

$$H = k_1 s_H$$

Direct proportion is justified by

Therefore k_1 is the gradient of the graph of H against s_H (x, y).

$$H = e^{-k_2 N}$$

Therefore $-k_2$ is the gradient of $\ln(H)$ against N .

$$H = e^{k_3 P}$$

Therefore k_3 is the gradient of the graph of $\ln(H)$ against P .

Assuming independence of parameters, we get:

$$H^3 = k_1 s_H e^{k_3 P - k_2 N}$$

$$H = (k_1 s_H e^{k_3 P - k_2 N})^{\frac{1}{3}}$$

$$\% \text{ Homeless} = 100 * H * P^{-1} = 100 * P^{-1} (k_1 s_H e^{k_3 P - k_2 N})^{\frac{1}{3}}$$

- For the mean income per homeless person, we use interpolation on the income percentiles.
- $(\text{Median-LB})/(\text{UB-LB}) = (\text{Median Position} - \text{CF of class before median}) / (\text{CFreq of Class of Median} - \text{CF of Class before}) \rightarrow \text{Let } F=100$
- From our global assumption: $\% \text{ homeless} = \% \text{ lowest income}$.

In the following graphs, we consider how question 3:

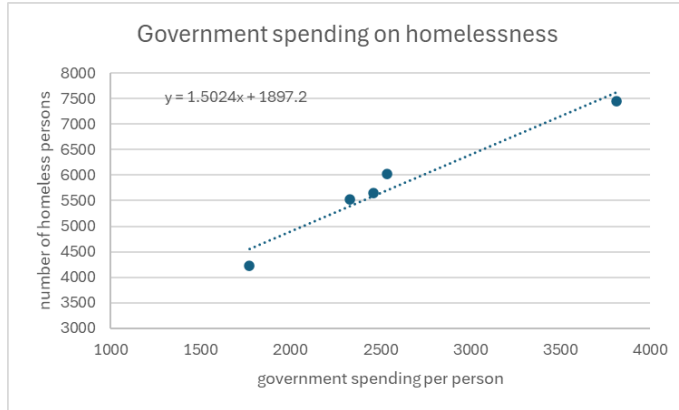
Manchester:

$$k_{1\text{Manchester}} = 1.5024$$

Brighton and Hove

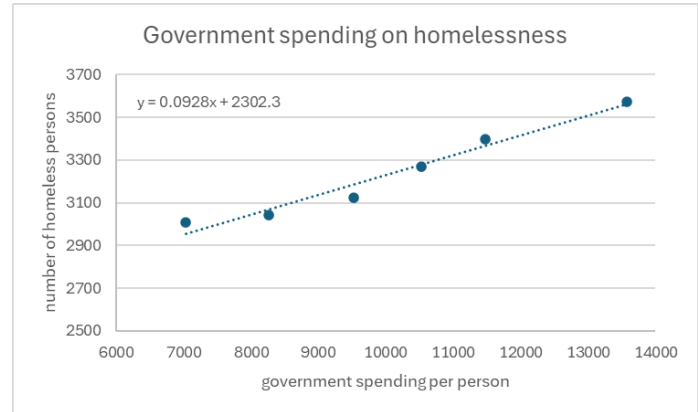
$$k_{1\text{Brighton and Hove}} = 0.0928$$

$$R^2 = 0.9508$$



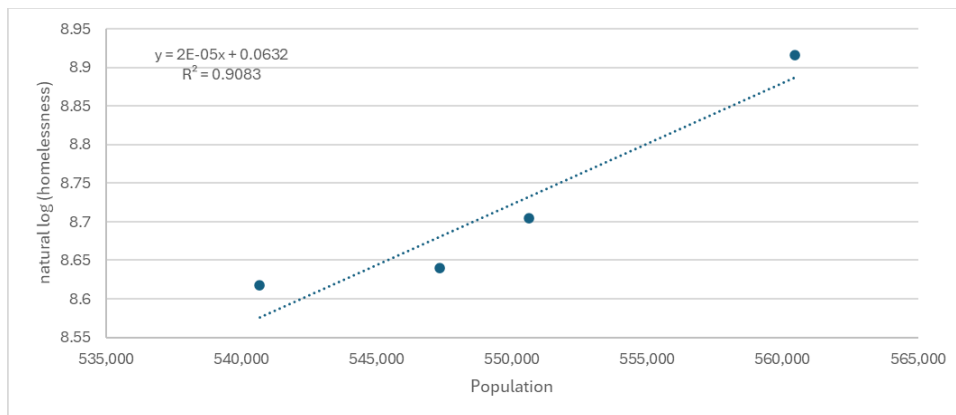
$$k_{1\text{Manchester}} = 1.5024$$

$$R^2 = 0.9636$$



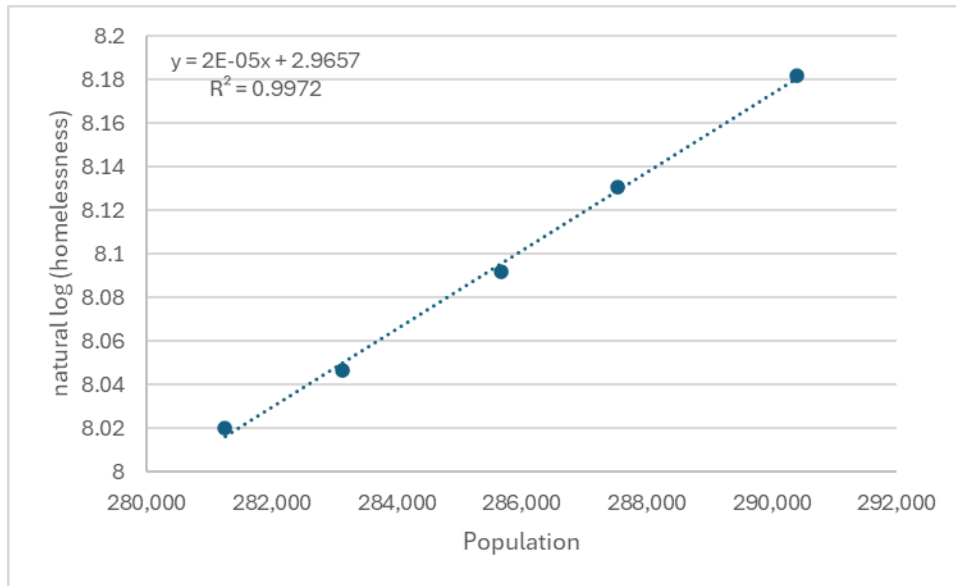
$$k_{1\text{Brighton and Hove}} = 0.0928$$

$$R^2 = 0.9508$$



Manchester

$$R^2 = 0.9636$$



Brighton

$$k_{2\text{Manchester}} = 6e^{-6}$$

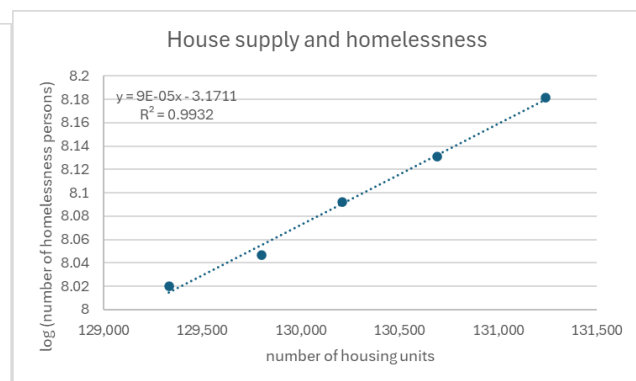
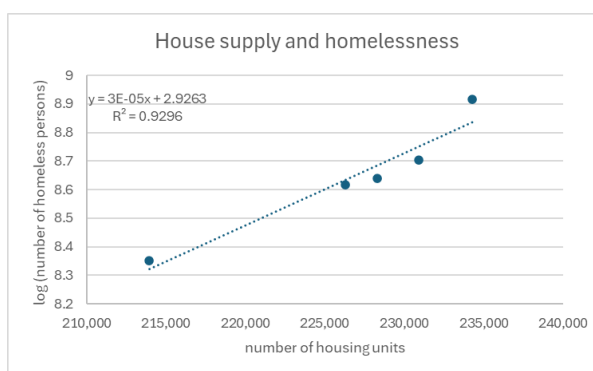
$$k_{2\text{Brighton and Hove}} = 5e^{-6}$$

$$R^2 = 0.8357$$

$$R^2 = 0.4586$$

Whilst the lower R^2 reflects the 2008/2009 financial crisis to an extent. As such low values aren't seen in the other four graphs it is important to consider the limitations of this graph.

Brighton



$$k_{3_{Manchester}} = 3e^{-5}$$

$$k_{3_{Brighton\ and\ Hove}} = 2e^{-5}$$

$$R^2 = 0.9296$$

$$R^2 = 0.4958$$

Question 2:

Therefore, we have a function for Manchester:

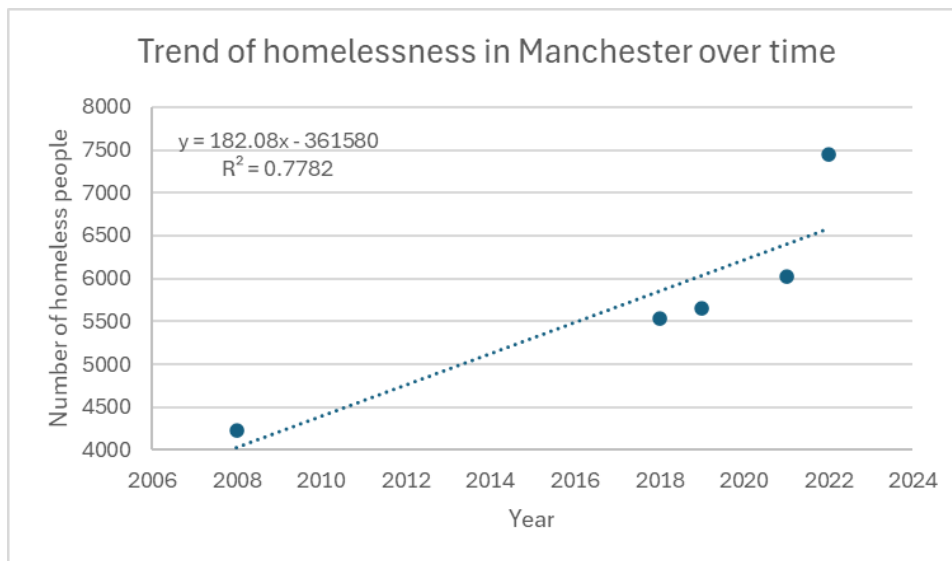
$$H_{Manchester} = (1.5024 * s_H * e^{3e^{-5}p - 6e^{-6}N})^{\frac{1}{3}}$$

And for Brighton & Hove:

$$H_{Brighton\ and\ Hove} = (0.0928 * s_H * e^{2e^{-5}p - 5e^{-6}N})^{\frac{1}{3}}$$

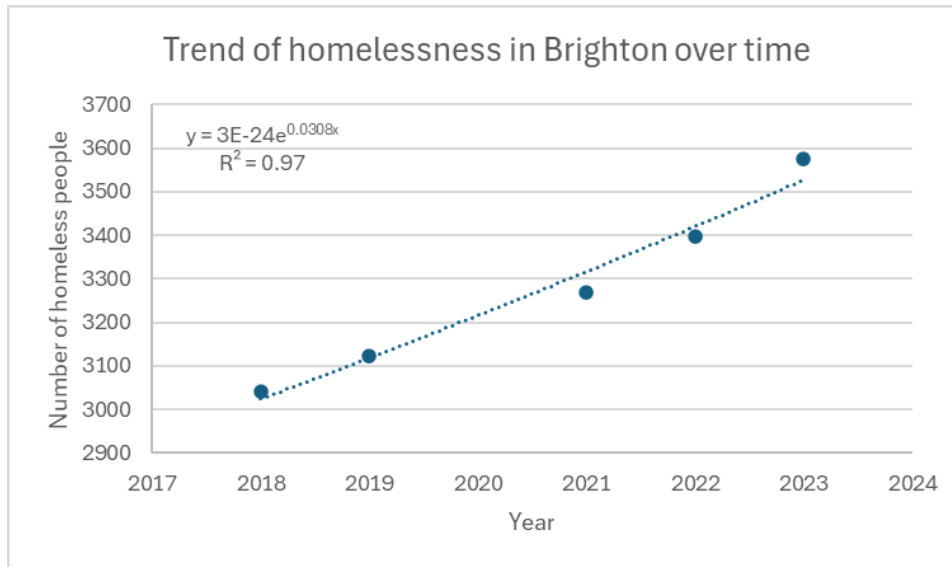
Testing

By comparing the function in the range of known data points (2008-2022)



We calculate an RMSE of 0.24. This value is relatively high. We suspect that this is a result of compounding uncertainties with exponents. With this we can extrapolate that the predicted

homeless number for 2033 will be 13200 and for 2043 it will be 23400 and for 2073 it will be 41200.



Using this we are able to extrapolate that our predicted number of homeless people for 2033 is approximately 4430, for 2043 it is approximately 4820 and for 2073 it is approximately 1130. We are able to do this because of the Granger causality explained below.

Choosing the Model

Our team utilized the Granger causality test to evaluate the significance of factors, identifying causal relationships between time series variables commonly in econometrics and finance. This test provides p-values for significance and F-statistic scores, with the F-test preferred for our continuous time-series data over the Chi-squared test, which is suited for categorical data. Our motivation being that we can reliably extrapolate trends, however due to covid there is a gap of 3 years with corrupted data, but by inspection and comparing the approximately equal gradients either side of the stationary point we concluded that it mealy acted as a lag, and therefore extrapolation is reasonable from this data set.

We will be using an α of 0.10 a measure of statistical significance, as this is standard practice for non-medical studies. 0.10 (10%): Indicates a 10% risk of concluding that a difference exists when there is no actual difference.

$$y_i = \alpha_0 + \sum_{j=1}^m \alpha_j y_{i-j} + \sum_{j=1}^m \beta_j x_{i-j} + \varepsilon_i$$

Figure 2: Granger causality general formula

The Granger causality test incorporates an additional factor called "lag," representing the time intervals required for causality to manifest. If, at a lag of 1, the p-value is below the threshold alpha value, it indicates a statistically significant cause-and-effect link within the dataset. Conversely, if at a lag of 15, the causality is not established (*p-value exceeds the alpha value*), this suggests that while there may be a significant correlation between the two datasets in the short term, this correlation does not extend to the long term. Our Threshold of Lag we have set at 4, as we want to see a long-term correlation (up to 50 years).

Executing the model:

In order to reliably extrapolate trends, we ensured that we used three different variables and looked at how effective they will be at predicting an outcome. We took all data available for the graphs, to ensure that even anomalies would be represented. After standardizing out time frame, we were able to integrate the number of homeless persons with another variable (*Population, Number of houses, Pounds spend per homeless person*). Then we employed the granger causality formula in the stats models using the python library as a combined data frame to access the Granger Causality test. This helped us understand what factors would influence Homelessness and look on what is causality.

Results:

The table consists of the factors tested with the three different lag values and their respective p-values and F-statistics for both cities. We checked whether they passed our maximum threshold or not (5% for α) and 4 for the Lag. We found that Manchester had a lot stronger causation, especially in the long term, however Brighton is struggling in short and long term, with some lags to great for us to consider it.

Factor	Lag (years)	F-statistics	p-value	Significance ($<\alpha *$)
Population (<i>Manchester</i>)	10	0.9203	0.0120	YES
	20	2.4035	0.0648	YES
	50	-	-	NO
Number of houses (<i>Manchester</i>)	10	0.8329	0.0323	YES
	20	0.7483	0.0920	YES
	50	3.2739	0.1040	NO
pounds spend per homeless person (<i>Manchester G.</i>)	10	2.3034	0.0127	YES
	20	0.9939	0.0098	YES
	50	2.0326	0.0102	YES

Factor	Lag (years)	F-statistics	p-value	Significance ($<\alpha *$)
--------	-------------	--------------	---------	------------------------------

Population (Brighton & Hove)	10	1.5682	0.0930	YES
	20	3.5898	0.3048	<i>NO</i>
	50	-	-	<i>NO</i>
Number of houses (Brighton & Hove)	10	1.0239	0.1130	<i>NO</i>
	20	3.9023	0.1327	<i>NO</i>
	50	-	-	<i>NO</i>
pounds spend per homeless person (Brighton & Hove)	10	1.2054	0.0024	YES
	20	0.8257	0.0342	YES
	50	0.0384	0.0102	YES

Table 5. Significance of factors with 10, 20, and 50 months of lag.

* α will be considered as 0.10

e. Discussion

The Granger causality test has given us insights into what factors most affect the number of homeless people, which include the number of houses, population, and pounds (of Government money) spent per homeless person. Our analysis showed that the causation for Manchester for those values was greater than for Brighton, however in the short term (10 years), all of them proved reliable. Interestingly, our results indicated that the number of houses available has a relatively low impact on homelessness in the long term for either city. This will help, as the focus for improving the statistics of homeless people will not be solved by just building more houses, but rather a combination of all three variables seen by the function plotted for question two. This can help for more focused and balanced investments to help reduce the number of homeless people.

f. Strengths & Weaknesses

In our model, we were able to use three functions to mitigate the effect of possible anomalies that occur by not sampling our data. This choice was made due to the fact that we wanted as pure of a representation as possible, knowing that reality will rarely behave as expected. This approach further allowed us to mitigate the effect of a disaster crisis happening in those cities, one example of this would be the financial crash in 2008, heavily affecting the data. However, with the 3 combined functions, we were able to get a good representation even with including these economic crises.

Integrating a function into three components, rather than relying solely on its mean, enables a more nuanced analysis of its behavior beyond mere input-output values. This approach significantly enhances comprehension of the underlying situation for both us and the reader. The application of the Granger causality test further substantiates this, revealing correlations

among variables and establishing causality, particularly with regard to the metric "expenditure per homeless individual." This evidence suggests that synthesizing these three aspects offers a comprehensive understanding of the situation's evolution across multiple factors. Moreover, the Granger causality test facilitates the extrapolation of our function to predict future values. This will also allow us to immediately use this model in question three,

However, one weakness of this model is that we did include outliers, which means that our trendline will have a greater margin of error. This affects the accuracy of the predictions that we can make with this model, which can be seen as some of our F-statistics are very bad, to the point that our program did not include them ($\text{Lag} > 4$).

3. Q3 Rising from This Abyss

a. Recomposing the question

This third task asks us to create a model that would address the for at least one of the two cities we have selected. We have chosen to draw a plan to fix the number of homelessness people in Manchester. The reasons underlying these decisions were that the data sample for Manchester were more important than the one for Brighton and Hove. Further, the linear models created to contest the previous two questions showed that R^2 was bigger in Manchester than in Brighton and Hove, while RMSC is more in Brighton and Hove than in Manchester. These finding allow us to create a model with greater confidence for the latter.

We define the success in addressing homelessness as follows. We need to limit the cost of the negative externality of homelessness on the community. Up until a certain threshold, homelessness costs more to the city than benefits. This model while explore the equilibrium between these two functions. Thus, the greatest amount of money saved will allow the community to invest in renewable energy or education.

Whilst some plans may seem more effective in the short term, we will consider a time frame of 20 and 50 years in the future, allowing us to make the long-term predictions relying on previously collected data.

We can reach the previously mentioned equilibrium by changing the variables used in task 2, relatively to their causation on homelessness – defined by the Granger Causality Test.

This model notably takes into account potential catastrophe – natural, financial, or migratory because it is adaptative, quantifying both the cost homelessness and homelessness benefits.

b. Assumptions

A – 1. We assume that the trend of number of homeless people stays linear in Manchester in the future if no different actions are taken from the actual ones.

Justification: It is an easy and straightforward way that allows us to determine a long-term plan to address homeless through modelling, as the same causes result in the same effects. We expect no departure from linearity because there hasn't been evidence of the contrary.

A – 2. We assume that the population can be controlled by increasing or decreasing immigration.

Justification: By controlling the number of immigrants allowed into the country, we ensure that there are no additions to the population from abroad therefore the population growth will only be subjected to the actual birth rate.

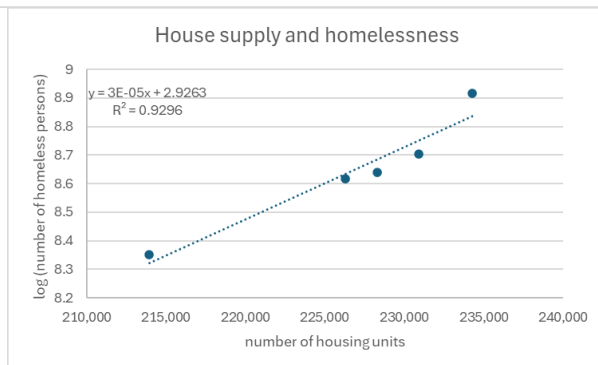
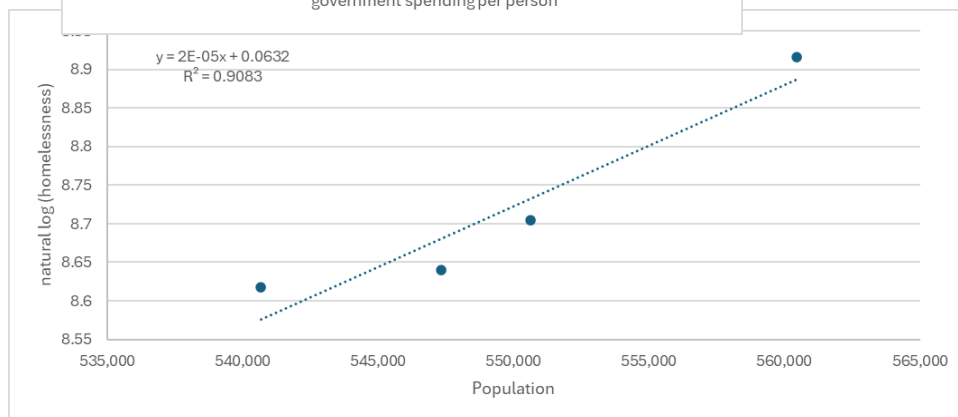
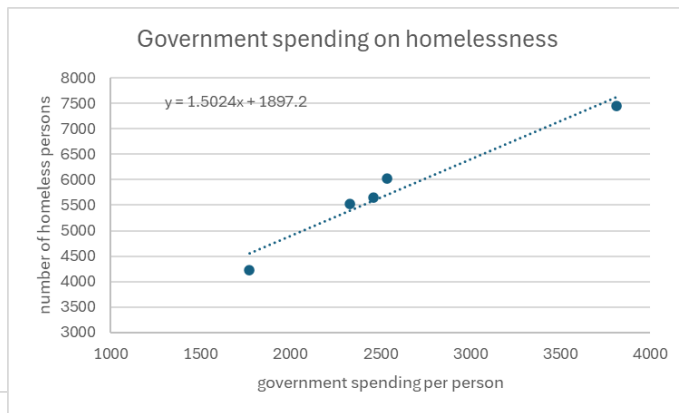
A – 3. We assume that the number of dwellings can be increased or decreased by increasing uses of fiscal policies, taxes and subsidies, as well as publicly funded housebuilding.

Justification: Government interventions can be used to directly increase or decrease the number of housing units by e.g. giving out subsidies to firms which will increase or decrease the cost of production, incentivizing firms to build more or less housing.

c. Variables

Symbols	Definition	Units
H	Number of homeless in a given city, per year	Units/year
N	Number of dwellings in a given city, per year.	Units/year
P	Number of inhabitants in a given city, per year	Units/year
s_H	s_T/H GBP of aid per homeless person	£/year

d. The Model



We could extrapolate

e. Strengths & Weaknesses

We acknowledge the need for further research which could make this final model more accurate. This could be realized notably by considering income and aid in conjunction, as this situation will differ from merely taking into account S_H .

One strength of our model is its depoliticization. It effectively lets the *invisible hand* to set the equilibrium, thus making it certain that utility will be maximized, especially in times of crises where the pitfalls created by unbalanced policies – of both austerity or overspending – can be avoided.

If given more time, we would start with a 3D heat map, where the mean y (cost per person) represent the mean cost per homeless person who becomes permanently housing.

Code Appendix

```

granger_cof.py
11 import pandas as pd
10 from statsmodels.tsa.stattools import grangercausalitytests
9
8 # Read the data from the spreadsheet
7 df = pd.read_excel('Downloads/Homeless_vs_Population_growth.xlsx', sheet_name='Sheet1')
6
5 # Select the columns of interest
4 data = df[['Data1', 'Data2']]
3
2 # Perform the Granger Causality Test (you also need to input the max lag to see till when you want to measure the value)
1 maxlag = 4
12 gc_test_result = grangercausalitytests(data, maxlag=maxlag, verbose=True)

```

```

from matplotlib import pyplot as plt
total_housing_units=''114,650
114,610
115,620
116,760
117,320
118,110
118,460
119,190
119,240
119,730
120,310
120,750
121,510
122,110
122,860
123,700
124,250

```

```

125,020
125,460
126,060
126,430
127,080
127,850
128,630
129,330
129,800
130,210
130,690
131,240'''

```

```

total_housing_units=total_housing_units.replace(",","")
total_housing_units=total_housing_units.replace("\n","")
housing_list=[]
temp=""
for char in total_housing_units:
    temp=temp+char
    if len(temp)==6:
        housing_list.append(temp)
        temp=""

for item in range(len(housing_list)):
    housing_list[item]=int(housing_list[item])

years=[]
for year in range(1993,2021+1):
    years.append(year)
plt.plot(years, housing_list)
plt.title("How the number of houses in Brighton changes over time")
plt.xlabel("Year")
plt.ylabel("Number of houses")
#plt.show()

print(years)
print(housing_list)

```

```

x=[1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004,
2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017,
2018, 2019, 2020, 2021]

```

```
y=[114650, 114610, 115620, 116760, 117320, 118110, 118460, 119190, 119240,  
119730, 120310, 120750, 121510, 122110, 122860, 123700, 124250, 125020,  
125460, 126060, 126430, 127080, 127850, 128630, 129330, 129800, 130210,  
130690, 131240]
```

```
NCap=linspace(1993, 2080, 88)
```

```
yext=interp1(x,y,NCap,'linear', 'extrap');
```

```
figure(1)
```

```
plot(NCap, yext, '--r')
```

```
hold on
```

```
plot(x,y)
```

```
hold off
```

```
grid
```

```
xlabel('year')
```

```
ylabel('number of houses')
```

```
#We repeated this method for the similar data set of both cities.
```

References

1. National Low Income Housing Coalition. "Zillow Research Examines Relationship Between Rent Affordability and Homelessness." <https://nlihc.org/resource/zillow-research-examines-relationship-between-rent-affordability-and-homelessness>
2. City Suites. "Manchester's Transport System Guide." <https://www.citysuites.com/en/blog/manchester-s-transport-system-guide/>
3. Brighton & Hove City Council. "Homelessness and Rough Sleeping Strategy 2020-2025." <https://www.brighton-hove.gov.uk/housing/homelessness-and-rough-sleeping-strategy-2020-2025>
4. Manchester City Council. "Manchester Housing Strategy 2022-2032." https://www.manchester.gov.uk/download/downloads/id/28615/manchester_housing_strategy_2022-2032.pdf
5. Brighton & Hove City Council. "Policies and Strategies - General Housing." <https://www.brighton-hove.gov.uk/housing/general-housing/policies-and-strategies-general-housing>
6. UK Government. "Coronavirus Guidance." <https://www.gov.uk/coronavirus#:~:text=There%20are%20no%20COVID%2D19,try%20to%20stay%20at%20home.>
7. UK Government. "Council Tax Stock of Properties 2021." <https://www.gov.uk/government/statistics/council-tax-stock-of-properties-2021>
8. Financial Times. "Article Title." <https://www.ft.com/content/32846f68-52fd-40e1-9328-ofe6bb3b9c19>
9. Trust for London. "Poverty Thresholds." <https://trustforlondon.org.uk/data/poverty-thresholds/#:~:text=The%20Minimum%20Income%20Standard%20identifies,housing%20costs%20for%20that%20year>
10. Brighton & Hove City Council. "Homelessness and Rough Sleeping Strategy 2020-2025." <https://democracy.brighton-hove.gov.uk/documents/s155601/Appx%201%20Homelessness%20Rough%20Sleeping%20Strategy.pdf>
11. Brighton & Hove City Council. "Homelessness and Rough Sleeping Strategy 2020-2025." <https://www.brighton-hove.gov.uk/housing/homelessness-and-rough-sleeping-strategy-2020-2025#:~:text=Brighton%20%26%20Hove%20Homelessness%20and%20Rough,end%20rough%20sleeping%20by%202027.>
12. Brighton & Hove City Council. "Homeless Strategy 2014." <https://www.brighton-hove.gov.uk/sites/default/files/migrated/article/inline/Homeless%20Strategy%202014.pdf>
13. Les Echos. "1875 euros mensuels de dépense publique par SDF." <https://www.lesechos.fr/2008/03/1875-euros-mensuels-de-depense-publique-par-sdf-1077962>
14. The Guardian. "From Social Care to Homelessness: What Are the Cost Pressures Facing English Councils?" <https://www.theguardian.com/uk-news/2024/jan/28/from-social-care-to-homelessness-what-are-the-cost-pressures-facing-english-councils>
15. Inside Housing. "Councils' Temporary Accommodation Spend Nears £1bn." <https://www.insidehousing.co.uk/news/councils-temporary-accommodation-spend-nears-1bn-57695>

16. The Argus. "Nearly £40 Million Spent on Temporary Housing for Homeless in Brighton." <https://www.theargus.co.uk/news/23915598.nearly-40-million-spent-temporary-housing-homeless-brighton/>
17. Manchester City Council. "The Budget 2019-20." <https://democracy.manchester.gov.uk/documents/s4612/The%20Budget%202019-20.pdf>
18. Greater Manchester Combined Authority. "Title of the Document." https://www.greatermanchester-ca.gov.uk/media/7870/230517_households_final.docx
19. Scholarpedia. "Granger Causality." [\[http://www.scholarpedia.org/article/Granger_causality\]](http://www.scholarpedia.org/article/Granger_causality)(
20. Numbeo. "Cost of Living Comparison between Brighton and Manchester." https://www.numbeo.com/cost-of-living/compare_cities.jsp?country1=United+Kingdom&city1=Brighton&country2=United+Kingdom&city2=Manchester
21. The Independent. "The Charts Which Show How Homelessness Has Soared Under the Conservatives." <https://www.independent.co.uk/news/uk/home-news/the-charts-which-show-how-homelessness-has-soared-under-the-conservatives-a7198966.html>
22. The Independent. "The Charts Which Show How Homelessness Has Soared Under the Conservatives." <https://www.independent.co.uk/news/uk/home-news/the-charts-which-show-how-homelessness-has-soared-under-the-conservatives-a7198966.html>
23. House of Commons Library. "Research Briefings: Homelessness in England." <https://commonslibrary.parliament.uk/research-briefings/cbp-7671/>
24. Mathematical Association of America. "Champion Team: Title of the Document." https://m3challenge.siam.org/wp-content/uploads/CHAMPIONTC-RUNNER-UP_16401.pdf